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Subject : Role and impact of Science and Technology in the development of India

Topic : Human diseases (Communicable and non- communicable)

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COMMUNICABLE DISEASES AND NON-COMMUNICABLE DISEASES

COMMUNICABLE DISEASES

The diseases which spread from one diseased person to another through contaminated food, water or contact or through insecticides, and animals are called the communicable diseases. These are caused by different causative agents (pathogens).

Diseases caused by viruses

1. Chicken pox

Pathogen: Chicken pox virus (varicella) Mode of transmission: By contact or through scabs Incubation period: 12-20 days

Symptoms

1. Fever, headache and loss of appetite
2. Dark red-coloured rash on the back and chest which spreads on the whole body. Later, rashes change into vesicles.
3. After few days these vesicles start drying up and scabs (crusts) are formed.
4. These scabs start falling (infective stage)

Prevention and cure

There are no vaccine against chicken pox as yet. But precautions must be taken as follows:

1. The patient should be kept in isolation.
2. Clothing and utensils, used by the patient should be sterilized.
3. Fallen scabs should be collected and burnt.
4. One attack of chicken pox gives lifelong immunity to the person recovered from this disease.

2. Measles

Pathogen: Virus (*Rubeola*)

Mode of transmission**:** By air

Incubation period: 3-5 days

Symptoms

1. Common cold
2. Appearance of small white patches in mouth and throat.
3. Appearance of rashes on the body.

Prevention and cure

1. The patient should be kept in isolation.
2. Cleanliness should be maintained.
3. Antibiotics check only the secondary infections which can easily recur.

3. Poliomyelitis

Pathogen : Polio Virus

Mode of transmissions : Virus enters inside the body through food or water.

Incubation period : 7-14 days

Symptoms

1. The virus multiplies in intestinal cells and then reaches the brain through blood.
2. It damages brain and nerves and causes infantile paralysis.
3. Stiffness of neck, fever, loss of head support.

Prevention and Cure

Polio vaccine drop (oral polio vaccine, OPV) are given to children at certain intervals.

Pulse polio programme is organized in our country to give polio vaccine to children

4. Rabies (also called hydrophobia)

Pathogen : Rabies virus

Mode of Transmission : Bite by a rabid dog.

Incubation period : 10 days to 1-3 months depending upon the distance of bite from Central Nervous System (CNS), that is the brain or spinal cord.

Symptoms

1. Severe headache and high fever.
2. Painful contraction of muscles of throat and chest.
3. Choking and fear of water leading to death.

Prevention and Cure

1. Compulsory immunization of dogs.
2. Killing of rabid animals.
3. Anti-rabies injections or oral doses are given to the person bitten by a rabid animal.

5. Hepatitis

Pathogen: Hepatitis B virus.

Mode of Transmission: Mainly through contaminated water.

Incubation Period: Generally, 15-160 days.

Symptoms

1. Body ache.
2. Loss of appetite and nausea.
3. Eyes and skin become yellowish, urine deep yellow in colour (due to bile pigments).
4. Enlarged liver.

Prevention and Cure

1. Hepatitis B vaccine is now available in India.
2. Proper hygiene is to be observed.
3. Avoid taking fat rich substances.

6. Influenza

Influenza, commonly known as 'flu' is an illness caused by viruses that infect the respiratory tract. Compared to common cold, influenza is a more severe illness.

Causes

Influenza is caused by a virus which attacks our body's cells, resulting in various effects depending on the strain of the virus.

There are many strains of influenza virus. The virus mutates all the time and new variations (strains) arise. This constant changing enables the virus to evade the immune system of its host. Unfortunately, immunity against one strain (which is conferred by exposure or immunization) does not protect against other strains. A person infected with influenza virus develops antibodies against that virus; as the

virus changes, the antibodies against the virus do not recognize the changed virus, and influenza can recur, caused by the changed or mutated virus.

Symptoms

Typical symptoms of influenza include:

1. Fever (Usually 100° F to 103° F in adults and often even higher in children).
2. Respiratory tract infection symptoms such as, cough, sore throat, running nose, headache, pain in the muscles, and extreme fatigue. Although nausea and vomiting and diarrhea can sometimes accompany Influenza infection, especially in children, gastrointestinal symptoms are rarely prominent.

Most people who get flu, recover completely in 1 to 2 weeks, but some people develop serious and potentially life-threatening complications, such as pneumonia.

Treatment and Control

1. Much of the illness and death caused by influenza can be prevented by annual influenza vaccination. Influenza vaccine is specifically recommended for those who are at high risk for complications with chronic diseases of the heart, lungs or kidneys, diabetes, or severe forms of anaemia.
2. The persons suffering from influenza should drink plenty of fluids take symptom relief with paracetamol, aspirin (not in children under the age of 16) or ibuprofen as recommended by the doctor. Consult doctor immediately for treatment.

7. Dengue

Dengue is an acute fever caused by virus.

It is of two types:

1. Dengue fever is characterized by an onset of sudden high fever, severe headache, pain behind the eyes and in the muscles and joints.
2. Dengue hemorrhagic fever is an acute infectious viral disease. It is an advanced stage of dengue fever. It is characterized by fever during the initial phase and other symptoms like headache, pain in the eye, joint pain and muscle pain, followed by signs of bleeding, red tiny spots on the skin, and bleeding from nose and gums.

Dengue spreads through the bite of an infected *Aedes aegypti* mosquito. The transmission of the disease occurs when a mosquito bites an infected person and subsequently bites a healthy person. In doing so, it transmits blood containing the virus to the healthy person and the person becomes infected with dengue.

The first symptoms of the disease occur about 5 to 7 days after the infected bite. *Aedes* mosquito rests indoors, in closets and other dark places, and is active during day time. Outside, it rests where it is cool and shaded. The female mosquito lays her eggs in stagnant water containers such as coolers, tyres, empty buckets, in and around homes, and other areas in towns or villages. These eggs become adults in about 10 days.

Incubation period

The time between the bite of a mosquito carrying dengue virus and the start of symptoms averages 4 to 6 days, with a range of 3 to 14 days.

Diagnosis

Diagnosis is made through blood tests by scanning for antibodies against dengue viruses. In addition, the blood platelet counts also get drastically reduced in the infected person.

Symptoms of Dengue fever

1. Sudden onset of high fever, generally 104-105 °F (40 °C), which may last 4-5 days.
2. Severe headache mostly in the forehead.
3. Pain in the joints and muscles, body aches.
4. Pain behind the eyes which worsens with eye movement.
5. Nausea or vomiting.

Symptoms of Dengue hemorrhagic fever

These include symptoms similar to dengue fever, plus other symptoms such as:

1. Severe and continuous pain in the abdomen.
2. Rashes on the skin.
3. Bleeding from the nose, mouth, or in the internal organs.
4. Frequent vomiting with or without blood.

5. Black stools due to internal bleeding.
6. Excessive thirst (dry mouth).
7. Pale, cold skin, weakness.

Prevention

Following steps can be taken to prevent spread of dengue fever:

1. Avoid water stagnation for more than 72 hours so that the mosquitoes do not breed there.
2. Prevent mosquito breeding in stored water bodies, like ponds, and wells.
3. Destroy discarded objects like old tyres and bottles, as they collect and store rain water.
4. Use mosquito repellents and wear long sleeved clothes to curtail exposure.
5. Use mosquito nets, also during daytime.
6. Avoid outdoor activities during dawn or dusk when these mosquitoes are most active.
7. Patients suffering from dengue fever must be isolated for at least 5 days.
8. Report to the nearest health centre for any suspected case of Dengue fever.

Treatment for dengue and dengue hemorrhagic fever

There is no specific treatment for dengue fever. Persons with dengue fever should rest and drink plenty of fluids. Dengue hemorrhagic fever is treated by replacing lost fluids. Some patients need blood transfusions to control bleeding.

Diseases caused by Bacteria

1. Tuberculosis

Pathogen: A bacterium (*Mycobacterium tuberculosis*).

Mode of Transmission: airborne-discharged through sputum, cough and sneeze, of the infected person.

Incubation period: 2-10 weeks during which the bacteria produce a toxin, tuberculin.

Symptoms

1. Persistent fever and coughing.
2. Chest pain and blood comes out with the sputum.
3. General weakness.

Prevention and Cure

1. Isolation of patient to avoid spread of infection.
2. BCG vaccination is given to children as a preventive measure.
3. Living rooms should be airy, neat and with clean sorroundings.
4. Antibiotics be administered as treatment.

2. Typhoid

Pathogen: A Bacillus rod-shaped bacterium (*Salmonella typhi*)

Mode of transmission: Through contaminated food and water

Incubation period: About 1-3 weeks

Symptoms

1. Continuous fever, headache, slow pulse rate.
2. Reddish rashes appear on the belly.
3. In extreme cases, ulcers may rupture resulting in death of the patient.

Prevention and Cure

1. Anti-typhoid inoculation should be given.
2. Avoid taking exposed food and drinks.
3. Proper sanitation and cleanliness should be maintained.
4. Proper disposal of excreta of the patient.
5. Antibiotics should be administered.

3. Cholera

It often breaks out among people in crowded areas and the areas with poor sanitary conditions.

Pathogen: Comma shaped bacterium (*Vibrio cholerae*)

Mode of transmission: Contaminated food and water. Housefly is the carrier.

Incubation period: 6 hours to 2-3 days.

Symptoms

1. Acute diarrhoea and watery stool.
2. Muscular cramps.
3. Loss of minerals through urine.
4. Dehydration leads to death.

Prevention and cure

1. Cholera vaccination should be given.
2. Electrolytes (Na, K, sugar) dissolved in water should be given to the patient to check dehydration (In market it is available as ORS–oral rehydration solution).
3. Proper washing and cooking of food.
4. Proper disposal of vomit and human excreta.
5. Flies should not be allowed to sit on eatables and utensils.

4. Diphtheria

This disease generally occurs in children of 1-5 years of age.

Pathogen: Rod-shaped bacterium (*Cornybacterium diphthereae*)

Mode of Transmission: Through air (droplet infection)

Incubation period: 2-4 days

Symptoms

1. Slight fever, Sore throat and general indisposition.
2. Oozing semisolid material in the throat which develops into a tough membrane.
3. The membrane may cause clogging (blocking) of air passage, resulting into death.

Prevention and cure

1. Immediate medical attention should be given.
2. Babies should be given DPT vaccine.
3. Sputum, oral and nasal discharges of the infected child should be disposed off.
4. Antibiotics may be given under doctor's supervision.
5. Isolation of the infected child.

5. Leprosy

Pathogen : A bacterium (*Mycobacterium leprae*)

Mode of transmission : Prolonged contact with the infected person. Nasal secretions are the most likely infectious material for family contacts.

Incubation period : 1-5 years

Symptoms

1. Affects skin.
2. Formation of nodules and ulcer.
3. Scabs and deformities of fingers and toes.
4. Infected areas lose sensation.

Prevention and Cure

1. The children should be kept away from parents suffering from leprosy.
2. Some medicine may arrest the disease and prevent from spreading.

Diseases caused by protozoans

1. Malaria

Pathogen : Malarial parasite (different species of *Plasmodium*)

Mode of transmission : By bite of female *Anopheles* mosquitoes

Incubation period : Approximately 12 days

Symptoms

1. Headache, nausea and muscular pain.
2. Feeling of chilliness and shivering followed by fever which becomes normal along with sweating after some time.
3. The patient becomes weak and anaemic.
4. If not treated properly secondary complications may lead to death.

Prevention and cure

1. Fitting of double door and windows (with "Jali" i.e. wire mesh) in the house to prevent entry of mosquitoes.
2. Use of mosquito net and mosquito repellents.
3. No water should be allowed to collect in ditches or other open spaces to prevent mosquito breeding.
4. Sprinkling of kerosene oil in ditches or other open spaces where water gets collected.
5. Antimalarial drugs to be taken.

2. Amoebiasis (Amoebic dysentery)

Pathogen: *Entamoeba histolytica*

Mode of transmission: Contaminated food and water

Symptoms

1. Formation of ulcers in intestine.
2. Feeling of abdominal pain and nausea.
3. Acute diarrhea and mucus in stool.

Prevention and cure

1. Proper sanitation should be maintained.
2. Vegetables and fruits must be properly washed before eating.
3. Antibiotics may be given to the patients.

Diseases caused by worms (helminths)

1. Filariasis

Pathogen : Filarial worm (*Wucheraria bancrofti*)

Mode of transmission : Bites of mosquitoes - *Aedes* and *Culex*.

Symptoms

1. Fever
2. Collection of endothelial cells and metabolites in the wall of lymph vessels.
3. Swelling takes place in certain parts of the body like legs, breasts, and scrotum.
4. Swelling of legs which appear as legs of elephant, so this disease is also called elephantiasis

Prevention and cure

1. Mesh doors and windows in the house to check the entry of mosquitoes.
2. The water collected in tanks or other articles should be properly covered.
3. Sprinkling of kerosene in ditches.
4. Drugs may be administered.

NON-COMMUNICABLE DISEASES:

Non-Communicable diseases do not spread from person to person. They are caused by other factors. Therefore, it is important to know which sickness are communicable and which are not. They are never caused by germs, bacteria, or other living organisms that infect the body. Antibiotics, or medicines that fight against germs do not help to cure non-communicable diseases.

Diabetes mellitus:

Diabetes mellitus is a chronic metabolic disorder. In Greek (Diabetes – running through; mellitus- sweet). It is characterized by increased blood glucose level due to insufficient, deficient or failure of insulin secretion. This is the most common pancreatic endocrine disorder. The incidence of Type-1 and Type-2 diabetes is increasing worldwide.

Type-1 Insulin Dependent Diabetes Mellitus (IDDM):

IDDM accounts for 10 to 20% of the known diabetics. The condition also occurs in children (juvenile onset diabetes) and young adults, the onset is usually sudden and can be life threatening. This is caused by the destruction of β-cells of the pancreas. It is characterized by abnormally elevated blood glucose levels (hyperglycemia) resulting from inadequate insulin secretion.

Causes: Genetic inheritance and environmental factors (infections due to virus, acute stress) are the cause for this condition.

Type-2 Non-Insulin Dependent Diabetes Mellitus (NIDDM):

This is also called as adult onset diabetes and accounting for 80 to 90% of the diabetic population. It develops slowly, usually milder and more stable. Insulin production by the pancreas is normal but its action is impaired. The target cells do not respond to insulin. It does not allow the movement of glucose into cells.

Causes: The causes are multifactorial which include increasing age (affecting middle aged and older people), obesity, sedentary life style, overeating and physically inactive.

Symptoms:

Diabetes mellitus is associated with several metabolic alterations. The most important symptoms are

1. Increased blood glucose level (**Hyperglycemia**).
2. Increased urine output (**Polyuria**) leading to dehydration
3. Loss of water leads to thirst (**Polydipsia**) resulting in increased fluid intake
4. Excessive glucose excreted in urine (**Glycosuria**)
5. Excess hunger (**Polyphagia**) due to loss of glucose in urine.

6. Fatigue and loss of weight

Prevention and Control of Diabetes

Diet, hypoglycemic drugs, insulin injection and exercise are the management options based on the type and severity of the condition. The overall goal of diabetes management is to maintain normal blood glucose level.

Dietary management:

Low carbohydrate and fiber rich diets are more appropriate. Carbohydrates should be taken in the form of starch and complex sugars. Refined sugars (sucrose and glucose) should be avoided. Diet comprising whole grains, millets (jowar, bajra, ragi), green leafy vegetables, wheat and unpolished rice should be included in diet regularly. Carbohydrates is maintained to about 50- 55% of the total calories. High protein content of 10-15% of the total intake is required to supply essential amino acids. Fat content in the diet should be 15-25% of the total calories. Saturated fat intake should be reduced. Polyunsaturated fatty acid content should be higher.

Management with insulin: Commercially available insulin preparations (short and long acting) are also used to maintain blood glucose levels. Physical activity: Exercise plays an important role in facilitating a good control of diabetes, in addition to strengthening and toning up the muscles. Education and Awareness: People with diabetics should be educated on the nature of disease they have and the possibility of complications of the disease, if blood sugar is not kept under control. Instructions regarding diet, exercise and drugs should be explained.

Heart Disease:

Cardiovascular disease (CVD) is associated with diseases of the heart and blood vessels. Coronary heart disease (CHD) is the most common form and is caused by deposition of cholesterol in the blood vessels. It usually develops slowly over many years beginning from childhood, they may form a fatty streak to a fibrous complicated plaque. It leads to the narrowing of blood vessels leading to atherosclerosis in the large and medium sized arteries that supply the heart muscle with oxygen. It leads to sudden ischemia (deficient blood supply to heart muscle) and myocardial infarction (death of the heart muscle tissue).

Risk factors: Hypercholesterolemia (High blood cholesterol) and high blood pressure (Hypertension) are the major causes and contributing factors for heart disease and if untreated may cause severe damage to brain, kidney and eventually lead to stroke. Causes: Heredity (family history), diet rich in saturated fat and cholesterol, obesity, increasing age, cigarette smoking, emotional stress, sedentary lifestyle, excessive alcohol consumption and physical inactivity are some of the causes. Symptoms: Shortness of breath, headache, tiredness, dizziness, chest pain, swelling of leg, and gastrointestinal disturbances.

Prevention and Control of Heart Disease

Diet management: Reduction in the intake of calories, low saturated fat and cholesterol rich food, low carbohydrates and common salt are some of the dietary modifications. Diet rich in polyunsaturated fatty acids (PUFA) is essential. Increase in the intake of fibre diet, fruits and vegetables, protein, minerals and vitamin are required.

Physical activity: Regular exercise, walking and yoga are essential for body weight maintenance

Addictive substance avoidance: Alcohol consumption and smoking are to be avoided.

Cancer:

Cancer causes about 4 million deaths annually throughout the world. In India more than one million people suffer from cancer. Cancer is derived from Latin word meaning crab. The study of cancer is called Oncology (Oncos- Tumor).

Cancer is an abnormal and uncontrolled division of cells that invade and destroy surrounding tissue forming a tumor or neoplasm (new growth). It is a heterogenous group of cells that do not respond to the normal cell division. The cancerous cells migrate to distant parts of the body and affect new tissues. This process is called metastasis. The frequent sites of metastasis are lungs, bones, liver, skin and brain.

Types of Cancers:

Cancers are classified on the basis of the tissues from which they are formed.

1. Carcinomas arise from epithelial and glandular tissues. They include cancers of skin, lung, stomach and brain. About 85% of the tumours are carcinomas.
2. Sarcomas are occurred in the connective and muscular tissue. They include the cancer of bones, cartilage, tendons, adipose tissue and muscles. These form 1% of all tumours.
3. Leukemia are characterized by an increase in the formation of white blood cells in the bone marrow and lymph nodes. Leukemia are called blood cancers. Most common type of cancer which also affect children below 15 years of age.

Carcinogenic Agents

Cancer causing agents are called carcinogens. They are physical, chemical agents, ionizing radiations and biological agents.

Physical Irritant: Heavy smoking causes lung cancer and cancers of oral cavity, pharynx(throat) and larynx. Betel and tobacco chewing causes oral cancer. Excessive exposure to sunlight may cause skin cancer.

Chemical agents: Nicotine, caffeine, products of combustion of coal and oil, pesticides, asbestos, nickel, certain dyes and artificial sweeteners induce cancer. Radiations: Ionizing radiations like X-rays, gamma- rays, radioactive substances and nonionizing radiations like UV rays cause DNA damage leading to cancer. Biological agents: Cancer causing viruses are called oncogenic viruses.

Treatment of Cancer:

The treatment of cancer involves the following methods:

Surgery: Tumours are removed by surgery to prevent further spread of cancer cells.

Radiation therapy: Tumour cells are irradiated by lethal doses of radiation while protecting the surrounding normal cells.

Chemotherapy: It involves administration of anticancerous drugs which prevent cell division and are used to kill cancer cells.

Immunotherapy: Biological response modifiers like interferons are used to activate the immune system and help in destroying the tumors.

Preventive measures for Cancer:

Cancer control programmes should focus on primary prevention and early detection. To prevent lung cancer tobacco smoking is to be avoided and protective measures to be taken against exposure to toxic pollutants of industries. Excessive exposure to radiation is to be avoided to prevent skin cancer.

Sexually Transmitted Diseases:

The diseases that are transmitted through sexual contact are known as sexually transmitted diseases. Sexually transmitted diseases are those diseases that are transmitted via the mucous membrane and secretions of the sexual organ, throat and the rectum. Syphilis, gonorrhoea, and AIDS are some sexually transmitted diseases.

AIDS (Acquired Immunodeficiency Syndrome)

AIDS is a severe viral disease and caused by Human Immunodeficiency Virus (HIV). It is a condition in which immune system fails and suppress the body's disease fighting mechanism. They attack the lymphocytes and the affected individual is prone to infectious diseases.

Transmission of HIV

AIDS virus has been found in urine, tears, saliva, breast milk and vaginal secretions. The virus is transmitted by an infected patient who comes in contact with blood of a healthy person. HIV/AIDS is not transmitted by touch or any physical contact. It spreads through contact of body fluids or blood. HIV is transmitted generally by

1. Sexual contact with infected person,
2. Use of contaminated needles or syringes especially in case of intravenous drug abusers.
3. By transfusion of contaminated / infected blood or blood products
4. From infected mother to her child through placenta.

Symptoms and Treatment of AIDS

Symptoms: Infected individuals become immunodeficient. The person becomes more susceptible to viral, bacterial, protozoan and fungal infections. Swelling of lymph nodes, damage to brain, loss of memory, lack of appetite and weight loss, fever, chronic diarrhea, cough, lethargy, pharyngitis, nausea and headache.

Diagnosis: The presence of HIV virus can be confirmed by Western Blot analysis or Enzyme Linked Immunosorbent Assay (ELISA)

Treatment: Anti-retroviral drugs and immune stimulative therapy can prolong the life of the infected person.

Prevention and Control of AIDS

The following steps may help in controlling and prevent the spreading of HIV infection

1. Screening of blood from blood banks for HIV before transfusion.
2. Ensuring the use of disposable needles and syringes in hospitals and clinics.
3. Advocating safe sex and advantages of using condoms.
4. Creating awareness campaign and educating people on the consequences of AIDS.
5. Persons with HIV/AIDS should not be isolated from the family and society.

Syphilis Causative organism

Treponema pallidum (a long corkscrew bacteria)

Mode of spread: Sexual contact with the infected person

Incubation period: Symptoms of the disease occur in about 10-90 days after contraction, but generally noticed in 3-4 weeks after getting infected with the bacteria.

Symptoms: Symptoms of syphilis occur in stages. The common symptoms of syphilis include.

1. Fever, and sores on the skin, in the throat and urinogenital areas especially vagina or penis, anus, rectum and mouth. Sores are firm, round and often painless.
2. Rashes on hands, feet and palms.
3. White patches in the mouth.
4. Acne-like warts in the groin area.
5. Hair fall in patches from infected areas.
6. The last three symptoms can be very serious.

They often become internal and affect organs like brain, nerves, liver, eyes, blood vessels, bones and joints, which show up after about 10 years of getting the infection. It can lead to paralysis, blindness, dementia and sterility.

Prevention and cure

1. Having sexual intimacy with only one person.
2. Avoiding prostitution and homosexuality.
3. Practicing abstinence, and use condoms.
4. Taking appropriate medical treatment, and maintain personal hygiene

Gonorrhea

Gonorrhea is a sexually transmitted disease that often involves urethra, vagina or penis, cervix, anus, and throat, as its target sites.

Causative organism: A gonococcus bacterium, *Neisseria gonorrhoeae*. It grows and multiplies quickly in warm moist areas of the body, such as the cervix, rectum and mouth.

Modes of spread: Having multiple sex partners increases the risk of contracting it. Any kind of unprotected sex is always a risk. Any kind of contact of sores with an infected person is also risky.

Incubation period: About 2-5 days after getting infection.

Symptoms

1. Inflammation of mucous membrane in the urinogenital tract.
2. Burning sensation while passing out urine and urethral discharge.
3. Rectal discomfort.
4. Pain in the joints.
5. Rashes on palms, Mild sore throat
6. In females, it may cause sterility

Prevention and cure

1. Having sexual contact with only one person.
2. Avoiding prostitution and homosexuality.
3. Taking antibiotics, such as penicillin injection or appropriate medicines at the appropriate time as per Doctor's advice.
4. In males, gonorrhea primarily affects the urethra, anus, throat, joints and eyes. Most victims of this disease are teenagers and young adults. One of the advanced complications of gonorrhea is gonococcal septicemia (blood poisoning).

Questions:

1. What are the diseases caused by Virus? Explain some of them.
2. Discuss in detail about the Diabetes mellitus.
3. Write a note on
   a) Cardiovascular disease
   b) Acquired Immunodeficiency Syndrome.